

Alternative-dependent diagnosability: a design-stage method for co-selecting system architecture and monitoring suite under a detectability requirement, with a demonstration on three wind-turbine generator topologies

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Abstract

Whether the failures that matter in service will be observable is rarely a criterion when the architecture of an engineered system is chosen; condition monitoring is specified later, for an architecture already fixed. This paper proposes a design-stage method that treats diagnosability as a property of the triple (architecture, control structure, observation suite) and evaluates it with one scale-free statistic, the signal-to-drift ratio: the relative change of an observable feature under a failure mode of known extent, divided by the relative change the healthy system exhibits on its own, pooled over all trials of the alternative. From it follow a minimum detectable extent and a detectable share for every combination of alternative, observation suite and failure-mode class; a six-step procedure co-selects architecture and monitoring suite under a detectability requirement and tests whether a monitoring design transfers across a product family. The method is demonstrated on three wind-turbine generator topologies subjected to identical winding short-circuits on open benchmark data. The same design-level fault produces different physical severities in the three machines; the control structure decides where the fault becomes visible; the cheapest observation suite meeting a requirement is architecture-specific, voltage transducers being decisive for one alternative and superfluous for another; and a learned detector transfers across architectures only when its features are referenced to the monitored unit's own healthy behaviour. The findings are robust to the method's tuning and carry recording-level uncertainty. Code reproducing every result from the public data is released.

Keywords: design for diagnosability, architecture selection, condition monitoring, sensor suite design, Design for X, wind-turbine generators, transferability, product family

1 Introduction

The architecture of an engineered system is chosen early, when little is known about how the system will fail, and it is chosen against criteria that are easy to quantify at that stage: performance, mass, cost, efficiency, manufacturability. Whether the failures that matter in service will be *observable*—and therefore detectable before they become expensive—is rarely among those criteria. Condition monitoring is instead specified later, by different people, for an architecture that is already fixed, and its difficulties are treated as signal-processing problems to be solved with better algorithms rather than as consequences of the design decision that created them. The Design-for-X literature has produced operational methods for manufacturability, assembly, reliability and maintainability, and, in electronics, for testability (Holt and Barnes, 2010; Simpson and Sheppard, 1994); for electromechanical systems, however, there is no equivalent method that lets a designer ask, at the point of choosing between architectures, “*which of these alternatives will let me see the faults I care about, with which sensors, and how early?*”

This paper argues that the question can be answered at the design stage, and gives a method for doing so. The central idea is to treat diagnosability as a property of the triple (architecture, control structure, observation suite) rather than of the fault, and to evaluate it with a single scale-free statistic that can be computed from any set of trials—measured or simulated—in which a failure mode of known extent is introduced into an otherwise healthy system. We call the statistic the signal-to-drift ratio (SDR): the relative change of an observable feature between a reference interval and a faulted interval, divided by the size of the relative changes that the same feature exhibits between adjacent reference intervals when nothing is wrong. An SDR of one marks the point at which a fault becomes distinguishable from the ordinary non-stationarity of the healthy system at a chosen per-trial false-alarm rate. From the SDR follow the two quantities a designer needs: the *minimum detectable extent* of a failure mode for a given alternative and observation suite, and the *share* of operating conditions in which that extent is detectable. Both are requirement-writable, both can be compared across alternatives whose signals live in different physical units, and both can be traded against the cost of the observation suite.

The motivating domain is the generator of a large wind turbine. The choice between a permanent-magnet synchronous generator (PMSG), a squirrel-cage induction generator (SCIG) and a wound-field synchronous generator (WFSG) is one of the established architecture-level trades in turbine design (Polinder et al., 2006, 2013), and electrical-subsystem failures are a major contributor to offshore downtime and unscheduled maintenance cost (Tavner et al., 2007; Carroll et al., 2016). Stator-winding insulation faults are the archetypal early-stage failure: they begin as a short-circuit across a few turns, produce subtle electrical signatures, and evolve into

machine-destroying phase faults if undetected (Gandhi et al., 2011; Riera-Guasp et al., 2015). A rich literature exists on detecting them in a given machine (Thomson and Fenger, 2001; Nandi et al., 2005; Bellini et al., 2008); almost none of it asks how the choice of machine changes the problem.

A recent series of open benchmark datasets makes that question answerable empirically for the first time. Tominaga et al. (2025a,b, 2024) subjected a PMSG, an SCIG and a variable-speed WFSG to the *same* controlled inter-turn and inter-winding short-circuits, inserted at the same tapped positions of the stator winding, at matched operating points, on instrumented laboratory benches; the PMSG and SCIG share a single bench and an identical test protocol. The datasets were published for algorithm benchmarking and explicitly refrain from comparing the machines. We use them for the opposite purpose: as the test-stage instantiation of a design method whose object is the comparison itself.

The contributions are:

1. A formal definition of *alternative-dependent diagnosability*, with the SDR statistic, its calibration against a pooled no-fault null, a reliability screen for observables, and the derived design quantities (minimum detectable extent, detectable share, signature location).
2. A six-step design procedure that uses these quantities to co-select a system architecture and its monitoring suite under a detectability requirement, and to assess whether a monitoring design will transfer across the alternatives of a product family.
3. A demonstration on three generator topologies under identical winding faults, yielding four empirical findings with design content: the same design-level fault is not the same physical fault across architectures; the control structure decides where in the drive the fault becomes visible; the cheapest adequate observation suite is architecture-specific; and a learned detector transfers across architectures only when its features are referenced to the monitored unit’s own healthy behaviour.
4. A robustness and uncertainty analysis of the method’s tuning choices (null quantile, window length) and of the recording-level variability behind every reported number, together with open code that reproduces all results from the public data.

Section 2 positions the work in the design and diagnosis literatures. Section 3 defines the method. Section 4 describes the three alternatives, the fault protocol, the observation suites and the computation. Section 5 reports the results, Section 6 draws out the design implications and the limits of what one machine per alternative can support, and Section 7 concludes.

2 Background and related work

2.1 Design-stage evaluation of lifecycle properties

Systematic design methodology treats the evaluation of alternatives against lifecycle criteria as a core activity of the conceptual and embodiment phases (Pahl et al., 2007; Tomiyama et al., 2009), and the Design-for-X programme has turned a growing list of such criteria into operational methods (Holt and Barnes, 2010). Reliability is the criterion closest to the present work. Failure mode and effects analysis remains the

dominant design-stage instrument; recent work in this journal has re-founded it on design theory (Cabanès et al., 2021), extended it with causal reasoning over failure data (Liu et al., 2026), embedded it in frameworks that demonstrate high reliability in regulated products (Booth and Bayford, 2026), and coupled it to probabilistic models of early design risk (Truong et al., 2026) and to machine-learning surrogates of degradation in robust design (Hassani and Khodaygan, 2026). What FMEA-type methods record about *detection* is, however, an expert rating; the “D” column of a DFMEA is assigned, not measured, and it is assigned for one architecture at a time. The method proposed here can be read as a way of measuring that column across alternatives.

The comparison of architectural alternatives is itself a design-methodological topic. Eaton et al. (2026) show that the technical measures omitted from an alternative-selection problem change which alternative is chosen, and Park and Okudan Kremer (2019) analyse how the topology of a product family constrains its evolution. Diagnosability is, we argue, a routinely omitted technical measure; and because monitoring designs are increasingly expected to serve a family of variants rather than one product (Simpson, 2004), whether a monitoring design transfers across alternatives is a family-level question that the selection method should answer.

2.2 Diagnosability, testability and sensor placement

Fault detection and isolation (FDI) has a mature theory of diagnosability for systems described by models. Structural analysis determines which faults can be detected and isolated from which sensor sets without numerical parameter values (Blanke et al., 2016; Krysander and Frisk, 2008; Frisk et al., 2009), and has been applied to electric-vehicle drives with a diagnosability index that ranks candidate residual generators (Zhang and Rizzoni, 2015). In electronics, testability analysis and multi-signal flow graphs answer the analogous question for test points (Simpson and Sheppard, 1994; Deb et al., 1994). Basseville (2001) gives statistical definitions of detectability and isolability for parametric models, and the prognostics community has developed sensor-selection criteria and performance metrics for health management (Cheng et al., 2010; Saxena et al., 2008). Model-based approaches (Isermann, 2005; Venkatasubramanian et al., 2003) require a model whose structure is known and trusted; data-driven fault diagnosis (Lei et al., 2020) requires data from the very system being designed, and its transfer across systems is an active problem (Pan and Yang, 2010).

The method proposed here occupies the space between these. It does not require a model structure, only trials with a known fault extent, which at the design stage can be generated by simulation and at the test stage by prototypes or benchmarks; it does not require a trained detector, only observable features; and it produces a quantity—the SDR—that is comparable across features, operating points and alternatives because it is normalised by the healthy system’s own non-stationarity rather than by a physical unit. It is closer in spirit to change-detection statistics (Basseville and Nikiforov, 1993) than to residual generation, and it deliberately stops at detectability: isolation and prognosis are downstream questions that presuppose it.

2.3 Condition monitoring of electrical machines

Inter-turn short-circuits in stator windings are among the most studied incipient faults in electrical machines (Gandhi et al., 2011; Riera-Guasp et al., 2015). Motor-current signature analysis (Thomson and Fenger, 2001), negative-sequence current and impedance methods (Kliman et al., 1996; Lee et al., 2003), transient models of turn faults (Tallam et al., 2002) and the review literature (Nandi et al., 2005; Bellini et al., 2008) have established the physical signatures: an asymmetry of the winding produces negative-sequence components in the terminal quantities and a component at twice the fundamental frequency in the synchronous reference frame. Which of these appears where depends on whether the machine is fed from the grid or from a current-controlled converter, because a current controller acts against the asymmetry it can see. This observation—well known to drive specialists—is the physical basis of one of our design findings, but it has not been turned into a design-stage measure. Machine-learning detectors, including our own recent work on multi-fault detection in PMSM drives, on permanent-magnet faults in offshore generators identified from finite-element simulations by transfer learning, and on onset-aware diagnosis (Canseven et al., 2026a,b; Sadık and Canseven, 2026), are typically developed and evaluated for one machine; the transferability experiments in this paper are the design-level counterpart of that line of work. Digital twins are the natural vehicle for the design-stage instantiation of the method (Tao et al., 2019); how such twins should themselves be designed is an open question in this journal (Tipuric et al., 2026; Tadeja et al., 2026).

2.4 Generator topology as a wind-turbine design decision

The generator concept of a large wind turbine—direct-drive or geared, permanent-magnet, induction or wound-field, full or partial power conversion—is selected early on cost, mass, efficiency and grid-compliance grounds (Polinder et al., 2006, 2013). Reliability enters the trade mostly through failure-rate statistics of subassemblies (Tavner et al., 2007; Carroll et al., 2016), not through the observability of the failures. Modern machine design optimisation has become highly effective at the electromagnetic and thermal objectives (Bramerdorfer et al., 2018; Canseven et al., 2024, 2025); the present method adds an objective that has so far been missing from that toolbox.

3 Method: alternative-dependent diagnosability

3.1 Definitions

Let $\mathcal{A}$ be a set of design alternatives. An alternative $a \in \mathcal{A}$ is an architecture together with the control structure it implies—for a generator, the machine topology and the converter control that the topology requires. Let $\mathcal{F}$ be a set of failure-mode classes; each class $f \in \mathcal{F}$ is parameterised by an *extent* $e \in E_f$ expressed at the design level, so that the same (f, e) can be realised in every alternative. For a winding fault, the extent is the fraction of a winding that is short-circuited; for a bearing it might be a defect size; for a structure a crack length. Let $\mathcal{S}$ be a set of candidate observation suites; a suite $s \in \mathcal{S}$ is a set of measurements with a cost $c(s)$, and $\Phi(s)$ is the set of

features that can be computed from those measurements alone. Let $\mathcal{O}$ be the operating envelope, discretised into operating points o .

A *trial* is a record of the system in alternative a at operating point o containing a reference interval R in which the system is healthy and a faulted interval F in which failure mode (f, e) is present. Trials may come from a physical test (test-stage instantiation) or from a validated simulation (design-stage instantiation); the method is indifferent to their origin, and the two instantiations can be combined by using the physical trials to validate the simulated ones.

3.2 The signal-to-drift ratio

For a feature $\varphi \in \Phi(s)$ and a trial, let $\bar{\varphi}_R$ and $\bar{\varphi}_F$ be the feature's mean values over the reference and faulted intervals, and define the relative change under fault

$$\delta_\varphi = \frac{\bar{\varphi}_F - \bar{\varphi}_R}{|\bar{\varphi}_R|}. \quad (1)$$

The reference interval is split into two adjacent halves R_1, R_2 of the same duration as F , and the relative change between them,

$$\delta_\varphi^0 = \frac{\bar{\varphi}_{R_2} - \bar{\varphi}_{R_1}}{|\bar{\varphi}_{R_1}|}, \quad (2)$$

measures how much the feature moves when nothing is wrong: sensor noise, drift of the operating point, controller activity, thermal effects. Because every trial has a healthy reference interval, δ_φ^0 is available from every trial of alternative a , faulted or not, and its absolute value pooled over all N_a trials forms an empirical null distribution of ordinary non-stationarity for that alternative and feature. The signal-to-drift ratio is

$$\text{SDR}_\varphi = \frac{|\delta_\varphi|}{q_\alpha(|\delta_\varphi^0|)}, \quad (3)$$

where q_α is the α -quantile of the pooled null. With $\alpha = 0.95$, $\text{SDR}_\varphi \geq 1$ means that the change under fault exceeds what the healthy system produces on its own in 95 % of comparable intervals: the fault is detectable at a per-trial false-alarm rate of approximately $1 - \alpha$ by the simplest possible detector, a threshold on the relative change of one feature.

Four properties make the SDR suitable as a design measure. It is *scale-free*: a current ratio, a voltage harmonic and a controller output are compared on the same axis, so features from different suites and alternatives whose signals live in different units can be ranked together. It is *calibrated* by construction against each alternative's own non-stationarity, so an alternative with a noisier healthy behaviour is not credited with detectability it does not have. It is *per-trial*, so its distribution over trials carries the dependence on operating point and fault extent that the design quantities below summarise. And it requires *no detector*: it upper-bounds what a threshold detector on that feature can do, and it does so before any algorithm is chosen, which is what makes it usable at the design stage.

Two safeguards are part of the definition. First, features whose SDR exceeds one on healthy trials—trials in which the fault interval contains no fault—more often than a tolerance τ are flagged as *unreliable* for that alternative and excluded from the design quantities; with nine healthy trials per alternative in the demonstration we use $\tau = 0.2$, i.e. at most one false alarm. Second, the analyst chooses whether a suite is evaluated by one *fixed* feature (the feature with the highest median SDR over the trials of a failure-mode class, chosen once) or by the *oracle* feature (the best feature of each trial). The fixed variant is what a threshold detector would achieve; the oracle variant is an upper bound for a detector that combines the suite’s features. We report both.

3.3 Derived design quantities

From the SDR distribution over trials we define, for each alternative a , suite s and failure-mode class f :

- the *diagnosability map* $D_a(s, f, e, o)$: the SDR of the suite’s fixed feature at extent e and operating point o ;
- the *minimum detectable extent* $\text{MDE}_a(s, f) = \min\{e : \text{median}_o D_a(s, f, e, o) \geq 1\}$, the smallest extent whose median SDR over the operating envelope reaches one;
- the *detectable share* $\text{DS}_a(s, f)$, the fraction of faulted trials with $\text{SDR} \geq 1$;
- the *signature location*: the group of measurements (e.g. stator currents, terminal voltages, controller internals, mechanical quantities) whose best reliable feature attains the highest SDR in the largest share of trials.

MDE and DS are the quantities against which requirements can be written (“inter-turn faults of extent $\geq e_{\text{req}}$ shall be detectable over the operating envelope at a per-trial false-alarm rate $\leq 1 - \alpha$ ”), and the signature location is the quantity that tells the designer *why* an alternative behaves as it does.

3.4 Design procedure

The procedure (Fig. 1) has six steps.

1. **Frame.** Enumerate the alternatives $\mathcal{A}$, the failure-mode classes $\mathcal{F}$ with a design-level extent scale E_f that is realisable in every alternative, the candidate observation suites $\mathcal{S}$ with their costs, and the operating envelope $\mathcal{O}$.
2. **Generate trials.** For each (a, f, e, o) obtain trials with a healthy reference interval and a faulted interval. At the design stage this is a simulation task (a finite-element or circuit model of each alternative with the fault injected at extent e); at the test stage it is a prototype or benchmark experiment. Include healthy trials with the same timing so that the reliability screen can be applied.
3. **Compute.** Extract the features $\Phi(s)$ for all suites on sliding windows within the intervals, form the pooled null per alternative and feature, compute the SDR per trial, and screen unreliable features.
4. **Map.** Build $D_a(s, f, e, o)$ and summarise it into $\text{MDE}_a(s, f)$, $\text{DS}_a(s, f)$ and the signature location, for the fixed and the oracle variants.
5. **Select.** For each alternative, find the cheapest suite that meets the detectability requirement; compare the alternatives on cost, MDE and DS together with their

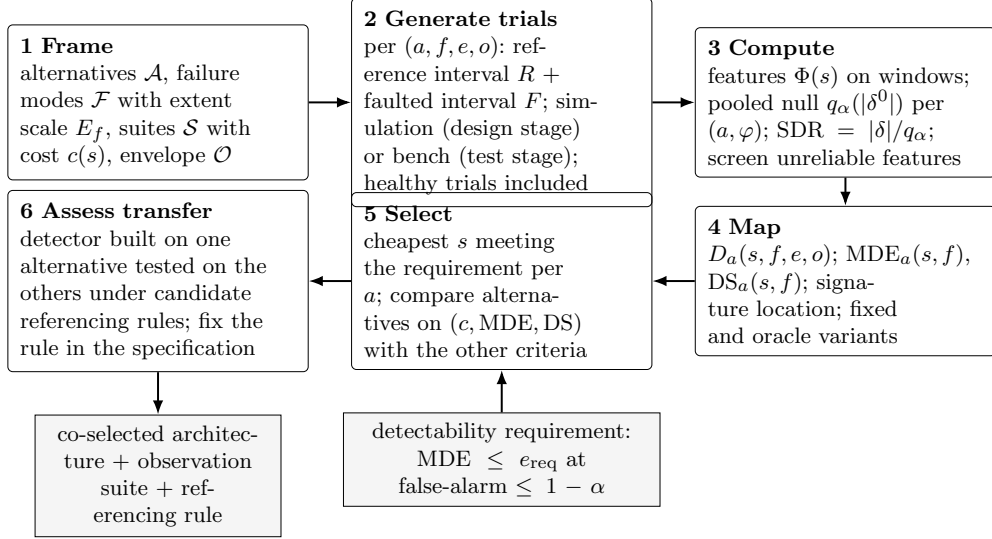


Fig. 1 The design procedure. Steps 3–4 are fully specified by Eqs. (1)–(3); steps 1, 2, 5 and 6 are design decisions the method structures.

other criteria. The output is a co-selection of architecture and observation suite, not a monitoring design for a given architecture.

6. **Assess transfer.** If the observation suite and its detector must serve a family of alternatives, test whether a detector built on one alternative’s trials detects the others’ faults, under the candidate feature-referencing rules (physical units; scaled on the target’s healthy trials; referenced to each trial’s own healthy interval). The referencing rule that transfers becomes part of the monitoring specification.

Steps 3–4 are mechanical and fully specified by the equations above; steps 1, 2, 5 and 6 are design decisions that the method structures but does not take.

3.5 Scope

The method evaluates detectability of a known failure-mode class at a known extent; it does not address fault isolation, severity estimation or remaining-life prediction, each of which presupposes detectability and can be built on the same trials. It also does not model the cost of missed detection; that enters through the requirement in step 5. Finally, the SDR is a property of a feature under a threshold detector: a detector that pools many windows or many features can detect below $\text{SDR} = 1$, which is why the oracle variant and the transfer experiment of step 6 are included.

Table 1 The three alternatives. PMSG and SCIG share one bench and one protocol; the WFSG bench differs where indicated.

	PMSG	SCIG	WFSG (variable speed)
Rating, poles	2.5 kVA, 4	2.5 kW, 4	2 kVA, 4 salient
Excitation	permanent magnets	induction (magnetising current)	wound field, controlled
Converter and control	2L, FOC with encoder	2L, FOC with flux observer	3L-NPC, FOC with encoder
Acquisition	20 kHz, 3 s	20 kHz, 3 s	4 kHz, 1.2–1.3 s
Pre-fault / fault duration	1.0 s / 0.40 s	1.0 s / 0.40 s	0.55 s / 0.12–0.27 s
Fault impedance	2.6 Ω	2.6 Ω	2.83 Ω (also 1–34 Ω)
Operating points	3 speeds $\times$ 3 torques	same	6 speeds, 4 torque levels (sparse)
Healthy trials	9	9	none
Fault trials used	216	216	198

4 Demonstration: three generator topologies under identical winding faults

4.1 Alternatives and data

The three alternatives are laboratory-scale generators of the same power class on instrumented test benches at the University of São Paulo, whose records were published as open benchmark datasets (Table 1). The PMSG (Tominaga et al., 2025a) and the SCIG (Tominaga et al., 2025b) are 2.5 kVA, four-pole machines from the same manufacturer, installed on the *same* bench: the same induction-motor prime mover and frequency converter, the same rapid-control-prototyping platform and field-oriented control (with an encoder for the PMSG and a flux observer for the SCIG), the same sensors, the same acquisition (20 kHz, 3 s records), the same fault-insertion hardware and the same test script. The variable-speed WFSG (Tominaga et al., 2024) is a 2 kVA four-salient-pole machine on an earlier bench of the same group with a three-level converter, field-oriented control with rotor-position measurement, 4 kHz acquisition and 1.2–1.3 s records. The PMSG–SCIG pair is therefore a controlled comparison in which everything but the machine and the control structure it requires is held fixed; the WFSG is a third alternative with the same fault protocol but a different bench, and is analysed with that difference kept explicit.

4.2 Failure-mode class and extent scale

All three machines were built with 24 tapped derivation points along their stator windings at the same relative positions, so that a short-circuit between two taps realises a fault of known extent (Table 2). Two classes are used: inter-turn short-circuits (both taps on the same winding) and inter-winding short-circuits (taps on the two parallel windings of a phase). The extent e is the fraction of the winding between the two taps; it ranges from 2.7 % to 12.1 % for the twelve inter-turn cases and from 2.3 % to 40.2 % for the twelve inter-winding cases, and it is identical across the three alternatives because the tap positions are. Faults were closed for about 0.4 s (PMSG, SCIG) or 0.12–0.27 s (WFSG) through a contactor and a fault resistance, after the machine had

Table 2 Tap pairs and fault extents (fraction of the winding between the taps), identical for the three alternatives.

Inter-turn			Inter-winding		
Taps	Phase	e (%)	Taps	Phase	e (%)
D11–D12	A	2.7	D04–D06	A	2.3
D09–D10	A	2.8	D16–D18	A	3.2
D23–D24	A	2.8	D04–D07	A	9.7
D21–D22	A	3.8	D16–D19	A	10.6
D06–D07	A	7.4	D01–D06	A	14.3
D14–D15	B	7.4	D13–D18	A	14.8
D18–D19	A	7.4	D01–D07	A	21.7
D02–D03	B	7.7	D13–D19	A	22.2
D17–D20	C	11.6	D12–D21	A	33.7
D13–D16	A	11.6	D11–D21	A	36.4
D01–D04	A	12.0	D12–D22	A	37.5
D05–D08	C	12.1	D11–D22	A	40.2

reached steady state; the fault current is measured but is used in this paper only as ground truth, never as a feature. The PMSG and SCIG matrix is complete: 24 fault cases $\times$ 3 speeds (1200, 1500, 1800 rpm) $\times$ 3 torques (5.2, 6.4, 8.0 N m), scaled from the operating region of the IEA 15 MW reference turbine (Gaertner et al., 2020), plus the healthy condition at each operating point. The WFSG matrix covers the same 24 cases at a sparser and unbalanced set of operating points with two to six repetitions each; we use the 198 trials at the fault impedance closest to the PMSG/SCIG value.

4.3 Observation suites and features

Five observation suites are considered, ordered by the hardware and integration they add to a bare generator–converter set (Table 7). *Drive-internal* uses only quantities the converter controller already has: the dq currents, its own dq voltage commands, the proportional–integral controller outputs, the modulation duty cycles, the DC-link voltage, the estimated electrical torque and the speed. *3 CT* adds three current transducers at the machine terminals and uses only angle-free features of the phase currents; *3 CT + 3 VT* adds three voltage transducers; *3 CT + 3 VT + drive* combines the external transducers with controller access, which also allows the measured terminal voltages to be transformed into the synchronous frame; and *+ torque transducer* adds a shaft torque sensor. For the WFSG the field current is an additional drive-internal observable (the *excitation* suite). Each feature is assigned to the cheapest suite whose measurements suffice to compute it; features that need the rotor or flux angle require controller access and are therefore never attributed to an external-transducer suite.

Features are computed on sliding windows of n_c electrical cycles with a hop of one cycle. The electrical frequency and the phase rotation are estimated per window from the phase slope of the current space vector, so that no angle sensor is needed for the external suites; sequence components are obtained by Hann-windowed projection of

Table 3 Observation suites and the features they can compute. f_e : electrical frequency; “ $2f_e$ ”: amplitude of the component at twice the electrical frequency; I_1, I_2, I_0 (V_1, V_2): positive-, negative- and zero-sequence magnitudes at f_e .

Suite	Added hardware / access	Features
Drive-internal	none; controller data	I_d, I_q : std, $1f_e, 2f_e$; $V_d^{\text{cmd}}, V_q^{\text{cmd}}$: $2f_e$; PI outputs d, q : std, $2f_e$; duty-cycle negative sequence D_2/D_1 ; T_e : std, $1f_e, 2f_e$; DC-link std; speed std
3 CT	3 current transducers	$I_2/I_1, I_0/I_1$, RMS unbalance, harmonics 3, 5, 7 of I_a relative to the fundamental, THD
3 CT + 3 VT	+ 3 voltage transducers	3 CT features + V_2/V_1
3 CT + 3 VT + drive	+ controller data	all of the above + measured V_d, V_q : $2f_e$
+ torque transducer	+ shaft torque sensor	+ T_m : std, $1f_e, 2f_e$
Excitation (WFSG)	field-current measurement	I_f : std, $2f_e$

each phase quantity onto the fundamental and symmetrical-component transformation; $1f_e$ and $2f_e$ amplitudes of synchronous-frame quantities by projection after mean removal. Appendix A lists the definitions.

4.4 Computation

For the PMSG and SCIG, the reference interval is 0.30–0.98 s after the start of each record (the contactor is commanded at 1.000 s), split at 0.62 s into R_1 and R_2 , and the faulted interval runs from 30 ms after the contactor command to 10 ms before its release; the fault current shows a relay pick-up delay of about 85–90 ms, so the faulted interval contains the insertion transient and the faulted steady state. For the WFSG the fault interval is located from the measured fault current (the first contiguous interval in which its 10 ms RMS exceeds a threshold set from the pre-fault noise), the reference interval is 0.05 s after the start of the record to 10 ms before the fault, split at 0.30 s. The main PMSG–SCIG comparison uses $n_c = 5$; the three-alternative comparison uses $n_c = 3$ for all three machines because the WFSG fault intervals are short, and the sensitivity of every conclusion to $n_c \in \{3, 5, 8\}$ and to $\alpha \in \{0.90, 0.95, 0.99\}$ is reported. The null is pooled over all 225 records of each machine (PMSG, SCIG) and all 198 records of the WFSG, every record contributing a healthy reference interval.

Uncertainty is quantified by bootstrap over trials (Efron and Tibshirani, 1993): 2000 resamples of the 108 trials of each alternative and fault class give percentile intervals for the median SDR, the detectable share and the ratio of median SDR between alternatives, and resampling the operating-point trials within each tap pair gives an interval for the MDE.

Step 6 of the procedure is instantiated as a window-level change detector: a gradient-boosted tree classifier (Lei et al., 2020) trained to separate faulted windows from healthy ones, evaluated within an alternative by five-fold cross-validation grouped by tap pair (so that the tested fault locations are never seen in training), and across alternatives by training on all trials of one machine and testing on all trials of another. Three feature-referencing rules are compared: physical units; features z -scored on the target machine’s healthy trials; and the absolute z -score of each window against the mean and standard deviation of its own trial’s reference interval. Negatives are the

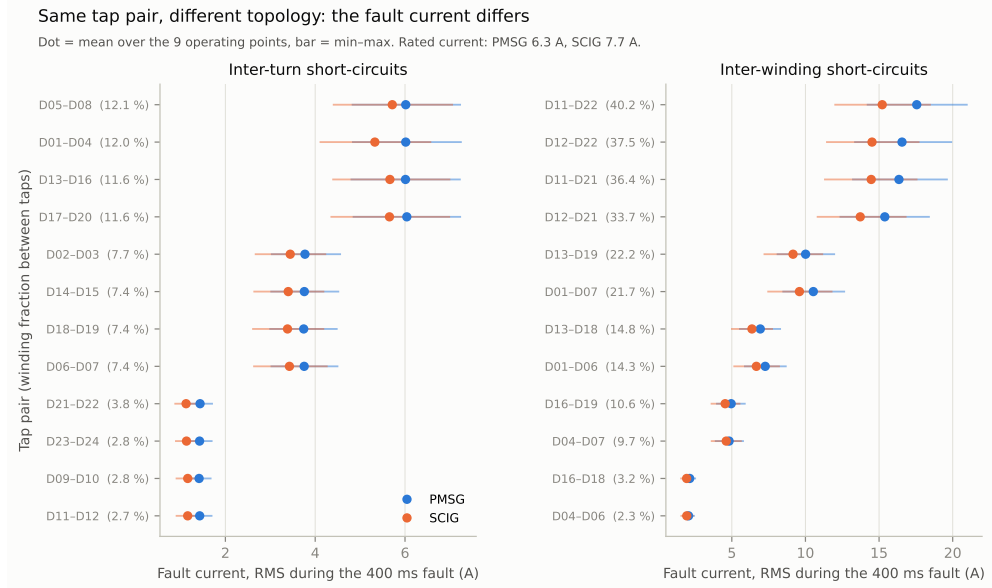


Fig. 2 Fault current per tap pair, PMSG and SCIG: mean over the nine operating points (dot) and range (bar). Rated currents 6.3 A (PMSG) and 7.7 A (SCIG).

reference windows of all trials plus the fault-interval windows of the healthy trials; the latter are the only negatives that contain the contactor operation without a fault and are therefore the informative ones.

All computations are implemented in Python and released with the paper; the pipeline runs from the public datasets to every table and figure in under fifteen minutes on a four-core machine.

5 Results

5.1 The same design-level fault is not the same physical fault

Figure 2 shows the fault current for each tap pair in the PMSG and the SCIG. For the same pair the SCIG’s fault current is 4–22 % lower (median ratio 0.91), but the difference that matters for detection is relative to what the stator sensors see: the fault current is 1.86 times the pre-fault stator current in the PMSG and 0.80 times in the SCIG, because the SCIG’s stator current carries the magnetising component (mean pre-fault current 7.1 A against 2.8–3.7 A). In per-unit of rated current the same pairs give 1.05 (PMSG) and 0.78 (SCIG). The WFSG trials at the matching impedance give fault-to-stator current ratios of 0.85 (inter-turn, 11.6–12.1 % extents) and 0.73 (inter-winding, 2.3–10.6 %), and 0.46 at the lower impedance used for its small inter-turn extents. A fault specified as “ e % of a winding” therefore arrives at the sensors of the three alternatives as three different physical events; Section 5.3 uses both the design-level and the physical axis.

Table 4 Null of the signal-to-drift ratio: 95th percentile of the absolute relative change of a feature between two adjacent healthy intervals, pooled over all trials of a machine ($N = 225$ for PMSG and SCIG, 618 for the WFSG), for five-cycle windows (PMSG, SCIG) and three-cycle windows (all three).

Feature	PMSG (5 c)	SCIG (5 c)	PMSG (3 c)	SCIG (3 c)	WFSG (3 c)
I_2/I_1	0.162	0.034	0.160	0.036	0.080
V_2/V_1	0.086	0.033	0.095	0.037	0.036
$I_d 2f_e$	0.563	0.016	0.444	0.015	0.158
$I_q 2f_e$	0.184	0.060	0.200	0.061	0.082
$PI_d 2f_e$	0.557	0.016	—	—	—
$V_q 2f_e$	0.165	0.060	0.176	0.061	0.096
$V_d^{\text{cmd}} 2f_e$	0.527	0.019	0.429	0.020	0.155
D_2/D_1	0.090	0.010	0.093	0.017	0.064
$I_a \text{ h3}$	0.216	0.035	0.162	0.039	0.560
$T_e 2f_e$	0.649	0.625	0.551	0.453	0.574
speed std	0.431	0.521	0.459	0.457	1.347
$I_f 2f_e$	—	—	—	—	0.339

5.2 Where the signature appears

Table 4 gives the null of the SDR—the 95th percentile of the healthy relative change—for the main features. It differs by up to a factor of thirty between machines: the SCIG’s synchronous-frame and current features are far quieter in relative terms than the PMSG’s, which is the first thing the method reveals and the first thing that must be kept in mind when the machines are compared (Section 6). Nine of the 31 features failed the reliability screen on at least one machine (e.g. the duty-cycle negative sequence on the SCIG fired on four of nine healthy trials, the speed standard deviation on the PMSG on four of nine) and are excluded from the design quantities below for that machine.

With the null in place, Fig. 3 shows the median SDR of every feature for the three alternatives, and Table 5 the share of trials in which each measurement group carries the strongest signature. The pattern is different in each machine. In the PMSG the terminal-voltage negative sequence V_2/V_1 is the strongest feature for both fault classes (median SDR 3.0 for inter-turn and 62 for inter-winding faults at five-cycle windows; 76 % and 94 % of trials detectable), followed by the duty-cycle negative sequence and the $2f_e$ component of the q -axis voltage; the stator-current negative sequence I_2/I_1 is weak for inter-turn faults (median SDR 1.5, 57 % detectable) and the controller group holds the strongest signature in 75 % and 90 % of trials. In the SCIG the strongest features are the $2f_e$ ripple of the d -axis current and of its PI output (6.7 and 74; 78 % and 100 % detectable), followed by the commanded d -axis voltage and by I_2/I_1 (4.5 and 56; 75 % and 100 %); the stator-current group is strongest in a third of the trials. In the WFSG the strongest inter-turn features are V_2/V_1 , the q -axis current ripple and the commanded q -axis voltage, and for inter-winding faults the q -axis ripple, the commanded voltage, the duty cycles and the third current harmonic; the field-current $2f_e$ ripple, an observable the other two machines do not have, detected 70 % of inter-turn and all inter-winding trials (Table B2). Two detectors trained on the PMSG and SCIG with the same features weight them almost orthogonally (cosine similarity of the logistic weight vectors -0.04).

Table 5 Signature location: share of fault trials (%) in which each measurement group’s best reliable feature attains the highest SDR (representative feature in parentheses). PMSG and SCIG, five-cycle windows.

Machine	Class	Stator currents	dq /controller	Mechanical
PMSG	inter-turn	16 (I_2/I_1)	75 (V_2/V_1)	9 (V_{dc} std)
PMSG	inter-winding	7 (I_2/I_1)	90 (V_2/V_1)	3 (V_{dc} std)
SCIG	inter-turn	33 (I_2/I_1)	62 ($PI_d 2f_e$)	5 (speed std)
SCIG	inter-winding	33 (I_2/I_1)	67 ($PI_d 2f_e$)	0 (speed std)

Table 6 Minimum detectable extent (MDE, % of the winding between the taps, bootstrap 95 % interval in brackets) and detectable share (DS, %) of four representative features, PMSG and SCIG, five-cycle windows, $\alpha = 0.95$. The smallest tested extents are 2.7 % (inter-turn) and 2.3 % (inter-winding).

Fault class	Feature	PMSG MDE	PMSG DS	SCIG MDE	SCIG DS
inter-turn	V_2/V_1	2.8 [2.7, 3.8]	76	2.7 [2.7, 7.4]	73
inter-turn	I_2/I_1	7.4 [2.7, 7.4]	56	2.8 [2.7, 7.4]	75
inter-turn	$PI_d 2f_e$	11.6 [3.8, 11.6]	31	2.7 [2.7, 7.4]	78
inter-turn	$V_q 2f_e$	7.4 [2.8, 7.7]	57	7.4 [7.4, 7.4]	60
inter-winding	V_2/V_1	2.3 [2.3, 2.3]	94	2.3 [2.3, 2.3]	100
inter-winding	I_2/I_1	9.7 [2.3, 9.7]	81	2.3 [2.3, 2.3]	100
inter-winding	$PI_d 2f_e$	10.6 [3.2, 14.3]	72	2.3 [2.3, 2.3]	100
inter-winding	$V_q 2f_e$	2.3 [2.3, 9.7]	90	2.3 [2.3, 9.7]	92

5.3 Minimum detectable extent

Table 6 gives the minimum detectable extent and the detectable share of four representative features, with their bootstrap intervals, and Fig. 4 plots the SDR against the physical severity axis for all three alternatives (the design-level axis is in Fig. B1 of the appendix). Through the terminal-voltage negative sequence both the PMSG and the SCIG detect inter-turn faults from the smallest tested extent (2.7–2.8 %) and inter-winding faults from 2.3 %. Through the stator-current negative sequence—the classic current-signature route—the SCIG still reaches the smallest extents, while the PMSG needs 7.4 % (inter-turn) and 9.7 % (inter-winding); through the d -axis controller ripple the PMSG needs 10.6–11.6 %. The SDR rises monotonically with extent in every machine (Spearman 0.69–0.95), and the bootstrap intervals of the MDE are wide because the tested extents form three clusters (about 3, 7.5 and 12 % for inter-turn faults), so the continuous SDR is the better basis for comparison: the ratio of median SDR between SCIG and PMSG is 3.1 [1.6, 5.6] for I_2/I_1 and 12.6 [5.0, 20.6] for the d -axis ripple on inter-turn faults, but 1.3 [0.5, 2.9] for V_2/V_1 and 2.0 [0.9, 3.2] for the q -axis voltage ripple (Table B5). The two machines differ where the signature relocates and are indistinguishable where it does not.

On the physical axis (Fig. 4) the picture is the same and independent of the tap-pair design: at equal fault-to-stator current ratio the SCIG’s current-side and d -axis features lie above the WFSG’s, which lie above the PMSG’s, while on V_2/V_1 the three machines nearly coincide. For the WFSG, whose small inter-turn extents were tested only at the lower impedance, the design-level MDE at the matching impedance is bounded by the smallest extent tested there (11.6 %, detectable in 78–82 % of trials);

Where the short-circuit becomes visible, for three generator topologies

3-cycle windows; ZF = 2.6 Ω (PMSG, SCIG) and 2.83 Ω (WFSG). Dot = median SDR, bar = interquartile range.

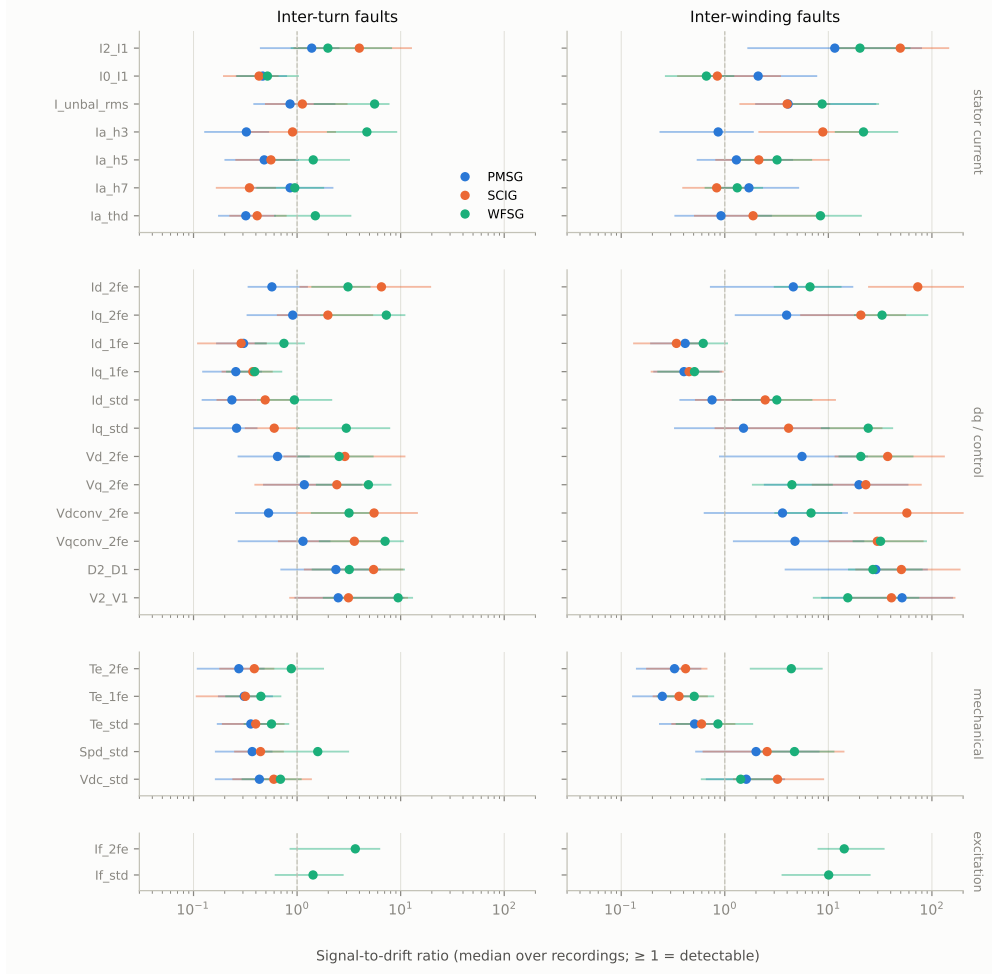


Fig. 3 Median signal-to-drift ratio of every feature for the three alternatives (dot) with interquartile range (bar), three-cycle windows; features to the right of the dashed line are detectable in more than half of the trials. The WFSG rows pool all its fault impedances.

its 2.7–7.7% inter-turn trials at 1Ω sit at $SDR \approx 1$ on V_2/V_1 and I_2/I_1 , and all its inter-winding extents from 2.3% are detectable.

5.4 Operating envelope

Speed dominates the operating-point dependence (Table B3, Fig. 5). In the PMSG the median SDR of the best feature rises from 5–9 at 1200 rpm to 15–20 at 1800 rpm and the detectable share from 79–83% to 92–96%; in the SCIG from 16–18 to 44–51 with 92–100% detectable throughout. Torque changes little. The dependence is

Detectability against physical fault severity, three topologies

Marker = fault impedance: $\blacktriangle$ 1 Ω , $\blacklozenge$ 2.6/2.83 Ω , $\blacksquare$ 5.66 Ω , $\blacklozenge$ 11.32 Ω (WFSG impedance rises with extent). Dashed = SDR 1.

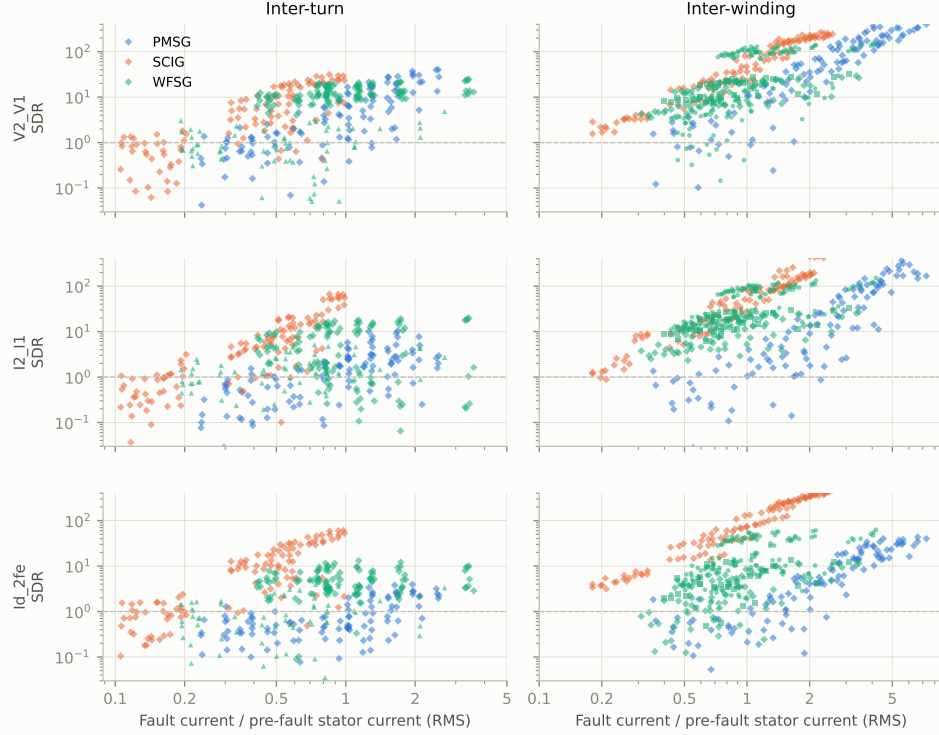


Fig. 4 Signal-to-drift ratio against physical fault severity (fault current relative to the pre-fault stator current) for three features and the three alternatives, all trials, three-cycle windows. Marker shape encodes the fault impedance of the WFSG trials.

consistent with the fault electromotive force scaling with speed, and it makes the lowest monitored speed a design parameter of the requirement.

5.5 Observation suites: what each level of hardware buys

Table 7 and Fig. 6 answer the design question for the PMSG and SCIG on five suites; Table B1 does the same for the three alternatives on the common feature set. For the PMSG, current transducers alone resolve inter-turn faults only from 7.4% and inter-winding faults only from 9.7%; adding three voltage transducers takes both to the smallest tested extent, and neither controller access nor a torque transducer improves the fixed-feature result further (controller access raises the oracle share from 88 to 95%). The drive-internal suite alone reaches 7.4% for inter-turn faults through the duty-cycle negative sequence, i.e. the converter’s own commanded quantities carry a weaker version of the signature that the measured terminal voltage carries (the commanded q -axis $2f_e$ component was excluded for firing on healthy trials). For the SCIG the cheapest suites already reach the smallest tested extents: current transducers

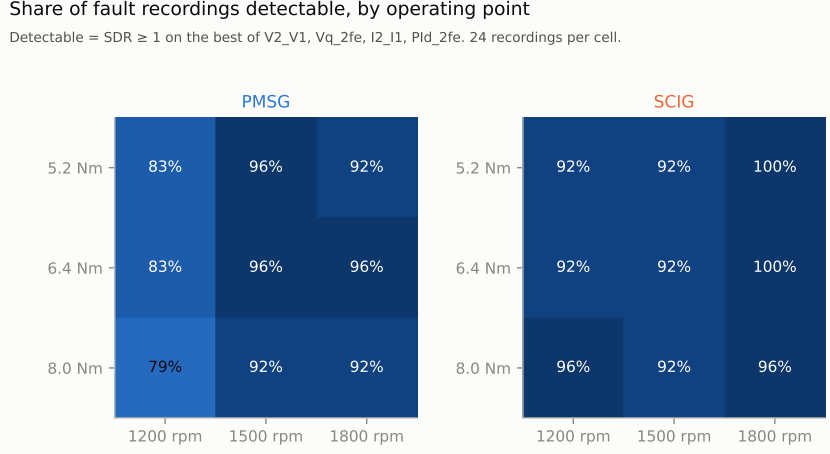


Fig. 5 Share of fault trials detectable per operating point, PMSG and SCIG (best of four features, five-cycle windows).

Table 7 Observation suites versus what they detect, PMSG and SCIG (five-cycle windows): best fixed feature of the suite, its minimum detectable extent (MDE, %) and detectable share (DS, %; per-trial oracle share in brackets). Features firing on more than one of the nine healthy trials of a machine are excluded for that machine.

Suite	PMSG feature	MDE	DS [oracle]	SCIG feature	MDE	DS [oracle]
<i>Inter-turn faults</i>						
drive-internal	D_2/D_1	7.4	64 [87]	$PI_d 2fe$	2.7	78 [96]
3 CT	I_2/I_1	7.4	56 [78]	I_2/I_1	2.8	75 [84]
3 CT + 3 VT	V_2/V_1	2.8	76 [88]	I_2/I_1	2.8	75 [88]
3 CT + 3 VT + drive	V_2/V_1	2.8	76 [95]	$PI_d 2fe$	2.7	78 [99]
+ torque transducer	V_2/V_1	2.8	76 [95]	$PI_d 2fe$	2.7	78 [99]
<i>Inter-winding faults</i>						
drive-internal	D_2/D_1	2.3	90 [100]	$PI_d 2fe$	2.3	100 [100]
3 CT	I_2/I_1	9.7	81 [94]	I_2/I_1	2.3	100 [100]
3 CT + 3 VT	V_2/V_1	2.3	94 [97]	I_2/I_1	2.3	100 [100]
3 CT + 3 VT + drive	V_2/V_1	2.3	94 [100]	$PI_d 2fe$	2.3	100 [100]
+ torque transducer	V_2/V_1	2.3	94 [100]	$PI_d 2fe$	2.3	100 [100]

through I_2/I_1 , and the drive-internal suite, with no added sensor, through the d -axis PI output; additional hardware raises the detectable share (75–78 % to 96–99 % oracle) rather than the reach. For the WFSG, whose inter-turn coverage is bounded as described above, the drive-internal, current-only and current-plus-voltage suites detect 78–82 % of inter-turn trials and all inter-winding trials from 2.3 %, and the field current alone detects 70 % and 100 %. A shaft torque transducer adds nothing for winding faults in either machine on which it was available (median SDR of the torque features 0.03–0.3).

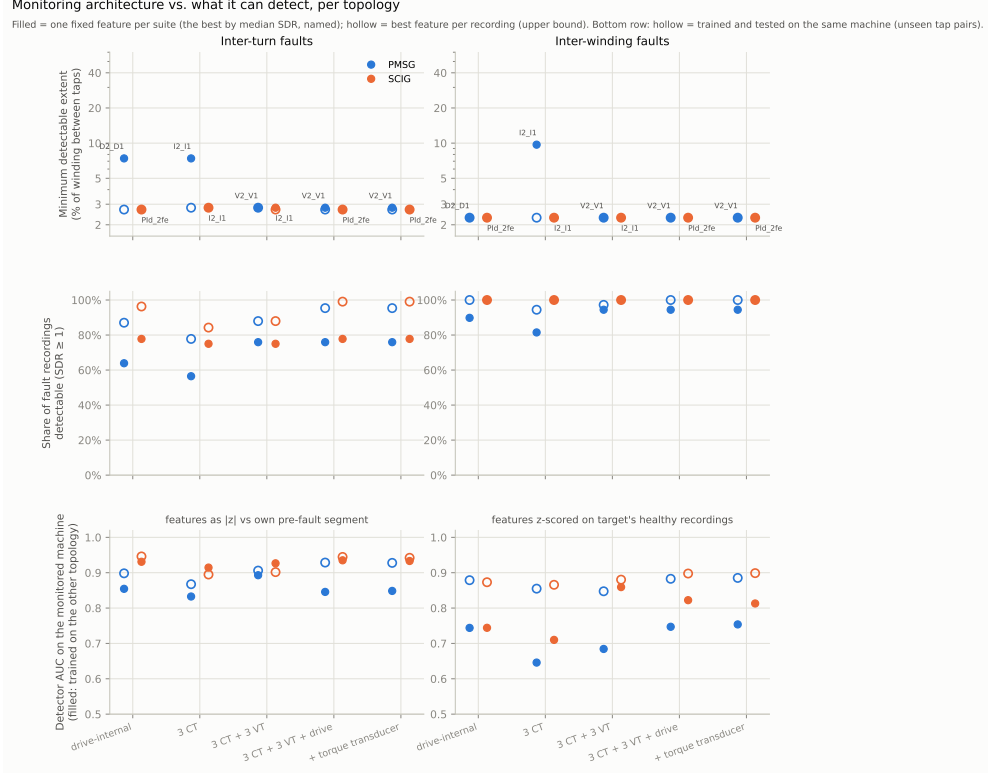


Fig. 6 Observation suites versus what they detect, PMSG and SCIG: minimum detectable extent (top) and detectable share (middle) for the fixed best feature (filled) and the per-trial oracle (hollow); bottom, cross-alternative transfer of a detector restricted to each suite’s features under two referencing rules (filled: trained on the other machine; hollow: trained on the same machine).

5.6 Transfer across alternatives

Table 8 reports step 6 for the PMSG–SCIG pair and Table 9 for the three alternatives. In physical units a gradient-boosted detector reaches an AUC of 0.89–0.90 within a machine on unseen tap pairs but 0.76 (PMSG→SCIG) and 0.59 (SCIG→PMSG) across; with its features scaled on the target machine’s healthy trials, 0.83 and 0.75; with each feature referenced to the trial’s own healthy interval, 0.93 and 0.85, against 0.93–0.94 within a machine. The corresponding true-positive rates at a 1% false-alarm rate are 0.35/0.28 (physical units), 0.23/0.31 (target-scaled) and 0.74/0.43 (self-referenced); a logistic detector on self-referenced features transfers with 0.91/0.89 AUC and 0.70/0.61 true-positive rate. Between the three alternatives, self-referenced detectors transfer in all six directions with AUC 0.81–0.98, and a detector trained on two machines detects the faults of the third with 0.84 (PMSG), 0.90 (SCIG) and 0.98 (WFSG); the WFSG entries are inflated by the absence of healthy trials on that bench, which leaves it without the nuisance-transient negatives the other two have. Restricted to each suite’s features (Fig. 6, bottom), the current-only suite transfers worst under

Table 8 Transfer of a window-level detector between PMSG and SCIG: AUC/true-positive rate at 1 % false-alarm rate. Within-machine columns use five-fold cross-validation grouped by tap pair; cross columns train on all trials of one machine and test on all trials of the other. Negatives are the reference and recovery windows of all trials and the fault-time windows of the healthy trials.

Referencing	Model	P→P	S→S	P→S	S→P
physical units	logistic	0.80/0.56	0.79/0.57	0.71/0.41	0.61/0.26
physical units	GBDT	0.89/0.61	0.90/0.65	0.76/0.35	0.59/0.28
z on target’s healthy trials	logistic	0.80/0.56	0.79/0.57	0.70/0.41	0.39/0.07
z on target’s healthy trials	GBDT	0.88/0.62	0.90/0.66	0.83/0.23	0.75/0.31
$ z $ vs own reference	logistic	0.91/0.71	0.92/0.80	0.91/0.70	0.89/0.61
$ z $ vs own reference	GBDT	0.93/0.76	0.94/0.83	0.93/0.74	0.85/0.43

Table 9 Transfer between the three alternatives (three-cycle windows, common features, $|z|$ against each trial’s own reference, GBDT). Same-machine rows: five-fold cross-validation grouped by tap pair. The WFSG has no healthy trials, so its negatives are reference and recovery windows only.

Trained on	Tested on	AUC	TPR at 1 % FPR
PMSG	PMSG	0.87	0.65
SCIG	SCIG	0.92	0.77
WFSG	WFSG	0.99	0.96
PMSG	SCIG	0.90	0.59
PMSG	WFSG	0.98	0.94
SCIG	PMSG	0.81	0.34
SCIG	WFSG	0.97	0.91
WFSG	PMSG	0.86	0.56
WFSG	SCIG	0.91	0.68
SCIG+WFSG	PMSG	0.84	0.42
PMSG+WFSG	SCIG	0.90	0.63
PMSG+SCIG	WFSG	0.98	0.94

target-scaling (0.65–0.71) and adequately under self-referencing (0.83–0.91); every suite transfers at 0.83–0.94 when self-referenced.

5.7 Robustness

Table B4 shows how the tuning choices move the design quantities. Lengthening the window from 3 to 8 cycles raises the median SDR by 10–20 % (longer averaging) and leaves every MDE of the PMSG unchanged; on the SCIG two inter-turn MDEs (V_2/V_1 , I_2/I_1) move from 2.7–2.8 % at five and eight cycles to 7.4 % at three, because those features sit just above the threshold at the smallest extents. Tightening the null quantile from 0.95 to 0.99 lowers the SDR by about a quarter and moves several inter-turn MDEs up one cluster; relaxing it to 0.90 does the reverse. Detectable shares move by at most nine percentage points across all settings, and no ordering of the alternatives on any feature reverses. The bootstrap intervals (Table B5, Fig. 7) show the recording-level variability behind each number: the shares are known to ± 8 –10 percentage points, the ratios of median SDR exclude one where the signature relocates and include it where it does not.

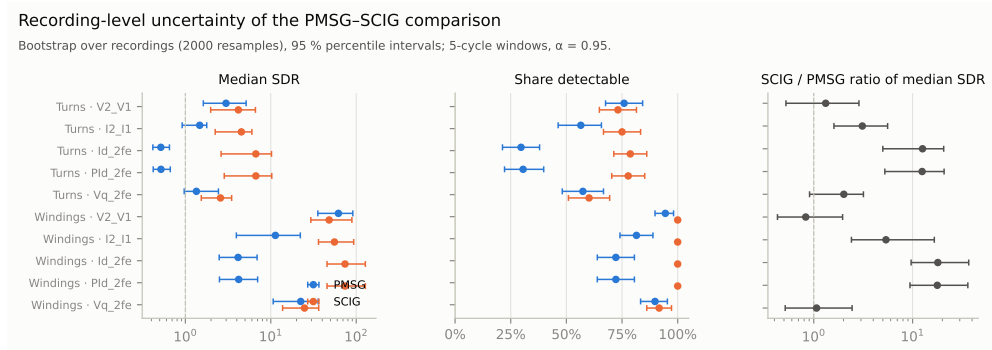


Fig. 7 Recording-level uncertainty: bootstrap 95 % intervals of the median SDR, the detectable share and the SCIG/PMSG ratio of median SDR for five features and both fault classes.

6 Discussion

6.1 What the demonstration establishes about the method

The method was proposed as a way of making diagnosability a criterion in architecture selection. The demonstration supports four claims about it that do not depend on which machine came out ahead.

The design-level description of a failure mode does not fix its physical severity. A short-circuit of the same winding fraction produced a fault current that differed by up to a factor of 1.3 between the PMSG and the SCIG and, relative to the current the stator sensors see, by a factor of 2.3 (Section 5.1). In the WFSG benchmark the experimenters had raised the fault impedance with extent, so that on that bench the extent axis and the impedance axis are confounded. Any design method that compares alternatives on a failure-mode class must therefore carry two severity axes—the design-level one, on which requirements are written, and a physical one, on which the alternatives are compared—and must state which one a number refers to. The SDR statistic makes this explicit, because it can be plotted against either (Figs. B1 and 4).

Where a failure becomes observable is decided by the control structure, not by the fault. With the fault physically similar, the PMSG’s current controller suppressed the current asymmetry and pushed it into the terminal voltage, whereas in the SCIG the asymmetry appeared in the magnetising-axis current and its controller action, and in the WFSG additionally in the field current and the current harmonics. The two detectors trained on the PMSG and SCIG weight the same features almost orthogonally. This is the mechanism behind the headline design finding: the same monitoring specification (“three current transducers, current-signature analysis”) gave a minimum detectable inter-turn extent of 7.4 % on the PMSG and 2.8 % on the SCIG, and adding three voltage transducers closed the gap on the PMSG but bought nothing on the SCIG. The cheapest adequate observation suite is a property of the alternative, and the reason can be read off the signature-location statistic.

A monitoring design transfers across alternatives only under the right referencing rule. A detector built on one machine and applied to another in physical units lost most of its performance; scaling its features on the target machine’s healthy trials recovered

part of it; referencing each feature to the trial’s own healthy interval recovered most of it, in all six directions between the three machines and when two machines were pooled to detect faults in the third. For a product family this is a specification rule, not an algorithmic detail: the monitoring function should be defined on deviations from the unit’s own reference behaviour, and the healthy trials that establish that reference are part of the commissioning procedure.

The measure is robust to its own tuning. The null quantile and the window length moved the reported quantities in the expected direction and by small amounts (Section 5.7); the bootstrap intervals separate the alternatives on the features where the signature relocates and overlap on the features where it does not. The method does not manufacture differences.

6.2 Design implications

For the wind-turbine generator, the results translate into statements a design team can use. A PMSG with full-power conversion should be specified with terminal voltage measurement if inter-turn faults below about 7% of a winding are to be detected by threshold monitoring; controller-internal quantities are not a substitute, because the commanded voltage carries a weaker and less reliable version of the signature than the measured one. An SCIG reaches the same detectability with current transducers alone or with no added sensor at all, and additional hardware buys robustness rather than reach. A wound-field machine offers a further drive-internal channel, the field current, whose $2f_e$ ripple detected every inter-winding fault and most inter-turn faults in the trials. A shaft torque transducer is not worth specifying for winding faults on any of the three. And a monitoring function meant to serve more than one generator type should be specified on self-referenced deviations.

More generally, the method gives DFMEA’s detection rating a measured basis: the “D” of a failure mode becomes a function of the alternative and the observation suite, with an uncertainty interval, rather than a consensus value. It also gives digital-twin efforts a concrete design-stage use: a twin that can inject failure modes of controlled extent and produce the same observables as the monitoring suite is exactly the trial generator of step 2, and its fidelity can be checked with the same statistic against a small number of physical trials, as the three benches were used here.

6.3 Applying the method elsewhere

Nothing in Section 3 is specific to electrical machines. What changes between domains is the content of the six framing decisions: what the alternatives are, what design-level extent scale is realisable in all of them, which suites are plausible, and what constitutes a trial. A gearbox family would use tooth-damage size as extent, vibration and oil-debris suites, and run-to-failure or seeded-fault trials; a battery-pack family would use cell-capacity loss, voltage/temperature/impedance suites, and cycling trials; a hydraulic actuator would use leakage cross-section, pressure/flow/position suites, and bench trials. The SDR, the pooled null, the reliability screen, the fixed/oracle distinction and the transfer test carry over unchanged. The one requirement the method imposes on the trial design—that every trial contain a healthy reference interval of at

least twice the faulted duration—is mild and, as the WFSG benchmark shows, is met even by data collected for other purposes.

6.4 Threats to validity

The most important limitation is that each alternative is represented by one physical machine. The differences reported are differences between three machines; attributing them to topology rather than to the individual units requires either replicated machines or, more realistically at the design stage, simulated populations of each alternative under parameter variation, validated against the measured units. We regard the demonstration as establishing the method and the mechanisms, not the population-level ranking of the three topologies. A related confound is the sensing chain: the SCIG’s magnetising current makes its stator current 2.5 times larger than the PMSG’s on the same current transducers, and its relative healthy drift is correspondingly smaller (Table 4); part of the SCIG’s advantage on current-side features may therefore be an effect of signal level on identical sensors rather than of topology. The method exposes this through the null table but cannot remove it without additional trials.

The benches are laboratory scale (2–2.5 kVA); the operating points were scaled from a 15 MW turbine, but the fault physics were not, and the fault durations were short (0.12–0.40 s). The fault impedance was fixed for the PMSG and SCIG and confounded with extent for the WFSG. Only winding faults were studied, and only electrical and controller observables; the method is not restricted to either, but its behaviour on vibration or thermal observables and on mechanical faults remains to be shown. The WFSG has no healthy trials, so its reliability screen rests on the pre-fault null alone, and its operating points are unbalanced. The transfer experiment’s negatives are dominated by reference and recovery windows; the fault-time windows of healthy trials, which contain the contactor operation without a fault, are the only negatives that resemble a nuisance transient, and other transients (load steps, grid events) were not in the data. Finally, the minimum detectable extent is a discrete quantity on a coarse extent grid; its bootstrap intervals are correspondingly wide, and the continuous SDR is the better quantity for comparing alternatives.

7 Conclusions

We have proposed a method that makes the diagnosability of failure modes a design-stage criterion for choosing between architectural alternatives and, jointly, for choosing the observation suite that will monitor the chosen architecture. Its core is a scale-free, self-calibrated statistic, the signal-to-drift ratio, from which a minimum detectable extent and a detectable share can be computed for every combination of alternative, suite and failure-mode class, and a six-step procedure that turns these into a co-selection and a transferability specification.

Applied to three wind-turbine generator topologies subjected to identical winding short-circuits on open benchmark data, the method showed that the same design-level fault is not the same physical fault across architectures, that the control structure decides where in the drive a fault becomes visible, that the cheapest observation suite meeting a given detectability requirement is architecture-specific—terminal voltage

transducers are decisive for the PMSG and superfluous for the SCIG—and that a learned detector transfers across architectures only when its features are referenced to the monitored unit’s own healthy behaviour. The findings are robust to the method’s tuning choices and are reported with recording-level uncertainty; they are limited by one physical unit per alternative, laboratory scale and a single failure-mode class.

Three extensions follow directly. First, the design-stage instantiation: simulated populations of each alternative under parameter variation, with the physical benches as validation points, would turn the demonstration’s machine-level differences into population-level statements and would let the method be applied before hardware exists. Second, other failure-mode classes and observables—bearing and rotor faults with vibration and flux observables—would test the method’s claim to generality within the same domain, and a mechanical or electrochemical family would test it across domains. Third, the cost side of the selection, treated here as an ordinal ranking of suites, deserves a proper model that includes integration, maintenance and the cost of missed detection, so that the co-selection can be posed as an optimisation. The code released with this paper reproduces every result from the public data and is intended as the starting point for those extensions.

Acknowledgements. The author thanks the InnovaPower group of the University of São Paulo for publishing the benchmark datasets on which this work rests.

Declarations

Funding. No specific funding was received for this work. [Author to complete: project or grant acknowledgements if applicable.]

Competing interests. The author declares no competing interests.

Ethics approval and consent. Not applicable; the work uses no human or animal subjects and no personal data.

Data availability. All data are public: the PMSG, SCIG and synchronous-generator datasets of the InnovaPower MitDev-Eletrica project (<https://github.com/InnovaPower/MitDev-Eletrica>; Zenodo DOIs 10.5281/zenodo.15741561, 10.5281/zenodo.17161986 and 10.5281/zenodo.13685630), licensed CC BY 4.0. The derived per-trial tables (features, signal-to-drift ratios, bootstrap samples) are provided with the code.

Code availability. The complete analysis pipeline (feature extraction, signal-to-drift computation, sensor-suite evaluation, transfer experiments, sensitivity analysis and all figures) is released under an open licence at [repository URL to be added at submission]; it reproduces every number in this paper from the public data.

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author’s direction as an assistant for code development, literature retrieval, data analysis and drafting and editing of the text. The author designed the study, verified

all analyses, checked every reference against its published record, and takes full responsibility for the content. No AI-generated images are used; all figures are produced by the released code from the data.

Appendix A Feature definitions

Let x_a, x_b, x_c be the sampled phase quantities of a window of N samples at sampling frequency f_s , w the Hann window and f_e the electrical frequency of the window. The complex fundamental amplitude of a signal x at frequency f is

$$X(f) = \frac{2}{\sum_n w_n} \sum_{n=0}^{N-1} x_n w_n e^{-j2\pi f n / f_s}. \quad (\text{A1})$$

The electrical frequency is estimated from the slope of the unwrapped phase of the current space vector $\frac{2}{3}(i_a + \mathbf{a}i_b + \mathbf{a}^2i_c)$, $\mathbf{a} = e^{j2\pi/3}$, whose sign gives the phase rotation. Sequence magnitudes are $|X_0| = |X_a + X_b + X_c|/3$, $|X_+| = |X_a + \mathbf{a}X_b + \mathbf{a}^2X_c|/3$ and $|X_-| = |X_a + \mathbf{a}^2X_b + \mathbf{a}X_c|/3$ for an abc rotation (interchanged for acb), evaluated at $f = f_e$ for currents (I_1, I_2, I_0) , terminal voltages (V_1, V_2) and mean-removed duty cycles (D_1, D_2) . The RMS unbalance is $\max_k |I_{\text{rms},k} - \bar{I}_{\text{rms}}|/\bar{I}_{\text{rms}}$ over the three phases. Harmonic ratios are $|I_a(kf_e)|/|I_a(f_e)|$ for $k = 3, 5, 7$, and the THD is $\sqrt{\sum_{k=2}^{20} |I_a(kf_e)|^2}/|I_a(f_e)|$. For synchronous-frame and scalar quantities y (currents i_d, i_q ; measured and commanded voltages v_d, v_q ; PI outputs; torques; field current) the features are the standard deviation of $y - \bar{y}$ and the amplitudes $|Y(f_e)|, |Y(2f_e)|$ of the mean-removed signal. Speed and DC-link features are standard deviations. Windows are n_c electrical cycles long with a hop of one cycle.

Appendix B Supplementary figure and tables

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Detectability versus fault extent for three generator topologies

Small dots = recordings; large markers = median per tap pair and impedance ($\blacktriangle$ 1 Ω , $\blacklozenge$ 2.6/2.83 Ω , $\blacksquare$ 5.66 Ω , $\blacklozenge$ 11.32 Ω). 3-cycle windows. Dashed = SDR 1.

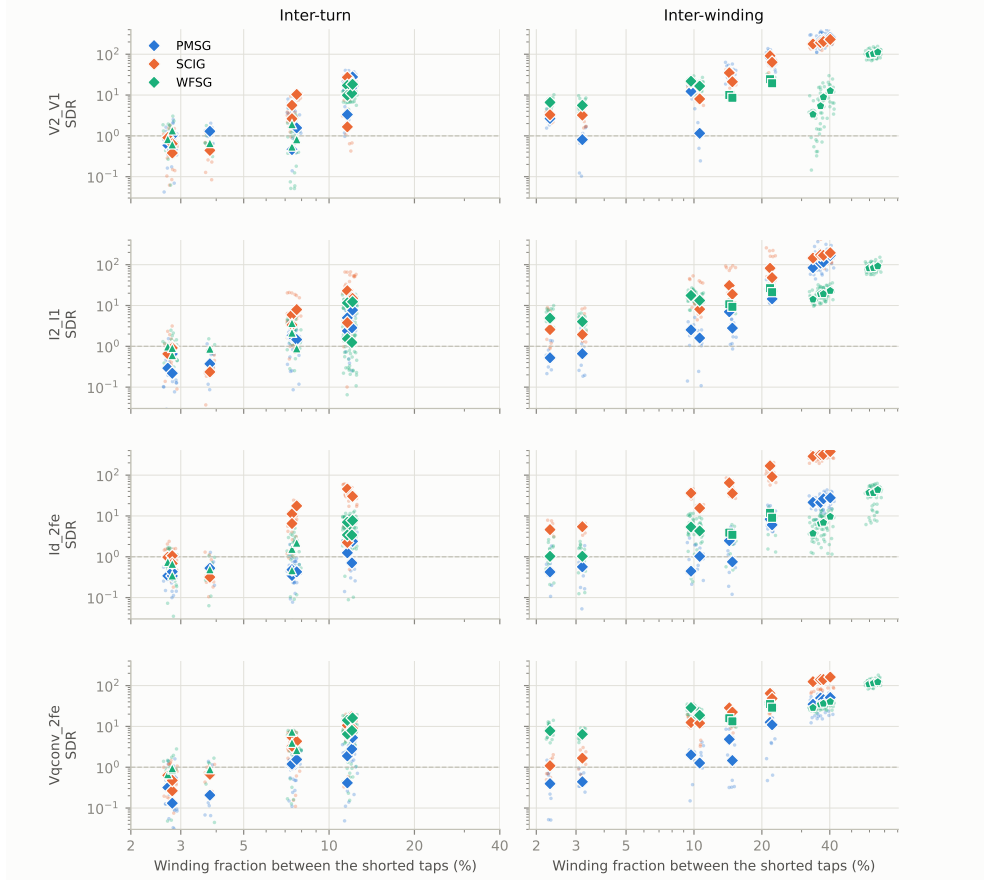


Fig. B1 Signal-to-drift ratio against the design-level fault extent for four features and the three alternatives (small dots: trials; large markers: median per tap pair and fault impedance, marker shape encoding the impedance), three-cycle windows.

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Table B1 Observation suites for the three alternatives, three-cycle windows, common feature set. MDE (%) at the fault impedance matching the PMSG/SCIG protocol; [†]the smallest inter-turn extent tested on the WFSG at that impedance is 11.6%, so the value is an upper bound. DS: detectable share (%) of the fixed feature; oracle: per-trial best feature.

Suite	Machine	Feature	MDE	DS	Oracle DS
<i>Inter-turn faults</i>					
drive-internal	PMSG	D_2/D_1	7.4	65	81
drive-internal	SCIG	$V_d^{\text{cmd}} 2f_e$	7.4	73	87
drive-internal	WFSG	$I_q 2f_e$	11.6 [†]	81	97
3 CT	PMSG	I_2/I_1	7.4	57	79
3 CT	SCIG	I_2/I_1	7.4	70	81
3 CT	WFSG	RMS unbal.	11.6 [†]	78	94
3 CT + 3 VT	PMSG	I_2/I_1	7.4	57	79
3 CT + 3 VT	SCIG	I_2/I_1	7.4	70	81
3 CT + 3 VT	WFSG	V_2/V_1	11.6 [†]	78	95
3 CT + 3 VT + drive	PMSG	D_2/D_1	7.4	65	92
3 CT + 3 VT + drive	SCIG	$V_d^{\text{cmd}} 2f_e$	7.4	73	94
3 CT + 3 VT + drive	WFSG	V_2/V_1	11.6 [†]	78	98
excitation (field current)	WFSG	$I_f 2f_e$	11.6 [†]	70	76
<i>Inter-winding faults</i>					
drive-internal	PMSG	D_2/D_1	2.3	91	98
drive-internal	SCIG	$V_d^{\text{cmd}} 2f_e$	2.3	100	100
drive-internal	WFSG	$I_q 2f_e$	2.3	100	100
3 CT	PMSG	I_2/I_1	9.7	83	99
3 CT	SCIG	I_2/I_1	2.3	98	98
3 CT	WFSG	$I_a \text{ h3}$	2.3	100	100
3 CT + 3 VT	PMSG	I_2/I_1	9.7	83	99
3 CT + 3 VT	SCIG	I_2/I_1	2.3	98	98
3 CT + 3 VT	WFSG	$I_a \text{ h3}$	2.3	100	100
3 CT + 3 VT + drive	PMSG	D_2/D_1	2.3	91	100
3 CT + 3 VT + drive	SCIG	$V_d^{\text{cmd}} 2f_e$	2.3	100	100
3 CT + 3 VT + drive	WFSG	$I_q 2f_e$	2.3	100	100
excitation (field current)	WFSG	$I_f 2f_e$	2.3	100	100

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Table B2 Five features with the highest median SDR per alternative and fault class (three-cycle windows, all WFSG impedances); detectable share in parentheses (%).

Machine	Class	Features: median SDR (DS)
PMSG	inter-turn	V_2/V_1 2.5 (73); D_2/D_1 2.4 (65); I_2/I_1 1.4 (57); $V_q 2f_e$ 1.2 (55); $V_q^{\text{cmd}} 2f_e$ 1.1 (55)
PMSG	inter-winding	V_2/V_1 51.5 (92); D_2/D_1 28.7 (91); $V_q 2f_e$ 19.8 (89); I_2/I_1 11.5 (83); $V_d 2f_e$ 5.6 (71)
SCIG	inter-turn	$I_d 2f_e$ 6.5 (78); $V_d^{\text{cmd}} 2f_e$ 5.5 (73); D_2/D_1 5.5 (78); I_2/I_1 4.0 (70); $V_q^{\text{cmd}} 2f_e$ 3.6 (66)
SCIG	inter-winding	$I_d 2f_e$ 73.0 (100); $V_d^{\text{cmd}} 2f_e$ 57.3 (100); D_2/D_1 50.7 (100); I_2/I_1 49.5 (98); V_2/V_1 40.7 (100)
WFSG	inter-turn	V_2/V_1 9.4 (80); $I_q 2f_e$ 7.3 (82); $V_q^{\text{cmd}} 2f_e$ 7.1 (82); RMS unbal. 5.6 (80); $V_q 2f_e$ 4.9 (81)
WFSG	inter-winding	$I_q 2f_e$ 33.0 (100); $V_q^{\text{cmd}} 2f_e$ 31.9 (100); D_2/D_1 26.9 (100); I_q std 24.3 (95); I_a h3 21.8 (100)

Table B3 Operating-point dependence: median SDR of the best of V_2/V_1 , $V_q 2f_e$, I_2/I_1 , $PI_d 2f_e$ and detectable share (%) per cell of the speed–torque grid, 24 fault trials per cell.

Machine	Speed (rpm)	5.2 N m	6.4 N m	8.0 N m
PMSG	1200	7.9 / 83	9.0 / 83	5.3 / 79
PMSG	1500	12.8 / 96	11.3 / 96	9.6 / 92
PMSG	1800	20.5 / 92	18.2 / 96	15.5 / 92
SCIG	1200	18.1 / 92	16.6 / 92	16.1 / 96
SCIG	1500	26.4 / 92	23.5 / 92	23.8 / 92
SCIG	1800	51.0 / 100	47.2 / 100	44.0 / 96

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Table B4 Sensitivity of MDE (%) / DS (%) to the window length (3, 5, 8 cycles at $\alpha = 0.95$) and to the null quantile ($\alpha = 0.90$ and 0.99 at five cycles). IT: inter-turn; IW: inter-winding.

Class	Feature	Machine	3 c	5 c	8 c	$\alpha 0.90$	$\alpha 0.99$
IT	V_2/V_1	PMSG	2.8/73	2.8/76	2.8/73	2.8/80	3.8/69
IT	V_2/V_1	SCIG	7.4/69	2.7/73	2.7/75	2.7/76	7.4/65
IT	I_2/I_1	PMSG	7.4/57	7.4/56	7.4/58	2.8/65	7.4/52
IT	I_2/I_1	SCIG	7.4/70	2.8/75	2.8/75	2.8/79	7.4/68
IT	$I_d 2f_e$	PMSG	11.6/29	11.6/30	11.6/28	11.6/33	11.6/24
IT	$I_d 2f_e$	SCIG	2.8/78	2.7/79	2.7/83	2.7/84	7.4/72
IT	$PI_d 2f_e$	PMSG	11.6/30	11.6/31	11.6/28	11.6/33	11.6/24
IT	$PI_d 2f_e$	SCIG	2.8/78	2.7/78	2.7/83	2.7/84	7.4/72
IT	$V_q 2f_e$	PMSG	7.4/55	7.4/57	7.4/60	7.4/65	7.4/50
IT	$V_q 2f_e$	SCIG	7.4/61	7.4/60	7.4/62	7.4/64	7.4/59
IW	V_2/V_1	PMSG	2.3/92	2.3/94	2.3/94	2.3/94	2.3/91
IW	V_2/V_1	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	I_2/I_1	PMSG	9.7/83	9.7/81	9.7/82	3.2/87	9.7/79
IW	I_2/I_1	SCIG	2.3/98	2.3/100	2.3/100	2.3/100	2.3/97
IW	$I_d 2f_e$	PMSG	10.6/71	10.6/72	10.6/69	10.6/74	14.3/69
IW	$I_d 2f_e$	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	$PI_d 2f_e$	PMSG	10.6/71	10.6/72	10.6/70	10.6/74	14.3/69
IW	$PI_d 2f_e$	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	$V_q 2f_e$	PMSG	2.3/89	2.3/90	2.3/91	2.3/91	9.7/86
IW	$V_q 2f_e$	SCIG	9.7/88	2.3/92	9.7/88	2.3/95	9.7/84

Table B5 Bootstrap over trials (2000 resamples, 95 % percentile intervals): median SDR per machine and ratio of medians SCIG/PMSG. Five-cycle windows, $\alpha = 0.95$. Intervals of the detectable shares are in Fig. 7.

Class	Feature	PMSG median SDR	SCIG median SDR	SCIG/PMSG
inter-turn	V_2/V_1	2.99 [1.62, 5.14]	4.17 [1.98, 6.61]	1.32 [0.52, 2.87]
inter-turn	I_2/I_1	1.47 [0.92, 1.78]	4.52 [2.24, 6.01]	3.10 [1.60, 5.59]
inter-turn	$I_d 2f_e$	0.52 [0.42, 0.65]	6.68 [2.62, 10.20]	12.61 [5.00, 20.64]
inter-turn	$PI_d 2f_e$	0.52 [0.42, 0.67]	6.69 [2.85, 10.27]	12.49 [5.24, 20.83]
inter-turn	$V_q 2f_e$	1.35 [0.97, 2.44]	2.58 [1.54, 3.48]	2.01 [0.91, 3.18]
inter-winding	V_2/V_1	61.90 [35.54, 91.27]	48.19 [29.43, 89.07]	0.83 [0.43, 1.96]
inter-winding	I_2/I_1	11.32 [3.95, 22.16]	55.61 [36.17, 93.45]	5.38 [2.41, 16.59]
inter-winding	$I_d 2f_e$	4.14 [2.49, 6.92]	73.97 [45.53, 127.88]	18.00 [9.66, 36.97]
inter-winding	$PI_d 2f_e$	4.21 [2.51, 7.01]	74.01 [45.55, 127.95]	17.78 [9.41, 36.22]
inter-winding	$V_q 2f_e$	22.40 [10.67, 35.86]	24.80 [13.75, 36.40]	1.07 [0.51, 2.44]

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